

Emotion Recognition in Conversation: A Comparative Study of Representation and Context Modeling

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Abstract

We study how different representation and contextual modeling strategies influence classification performance in emotion recognition over multi-turn dialogues, using the DailyDialog dataset. We compare two architectural approaches: an end-to-end model with partially fine-tuned RoBERTa at the utterance level, and a modular pipeline using frozen utterance embeddings—either generic or task-adaptive—as input to classifiers operating at the utterance or dialogue level. The best results are obtained with a dialogue-level Transformer encoder using frozen embeddings extracted from the fine-tuned RoBERTa model. Attribution analyses suggest that contextual information is encoded at the representation level but not consistently exploited in final predictions, indicating a gap between encoding and decision.

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1. Introduction

Emotion Recognition in Conversation (ERC) is a Natural Language Processing (NLP) task with applications in dialogue systems, mental health monitoring, and human-computer interaction (1). Unlike standard emotion classification, ERC involves short utterances whose emotional interpretation depends on dialogic context, speaker identity, and temporal dependencies. These factors make the task challenging, especially in the presence of class imbalance and semantic overlap among emotions.

This work evaluates two architectural strategies. The first is an end-to-end approach that performs utterance-level classification by partially fine-tuning a Transformer encoder with

a classification head. The second is a modular pipeline where precomputed utterance embeddings—either frozen or fine-tuned—are used as input to separate classifiers, operating either at utterance or dialogue level. This setup allows us to isolate the contributions of task-adaptive representations and context modeling.

Experiments are conducted on the DailyDialog dataset, with evaluation focused on non-neutral classes to assess the role of context and fine-tuning in disambiguating emotional content.

The rest of the report is organized as follows. Section 2 describes the dataset, embedding strategies, model architectures, and training protocol. Section 3 details the experimental setup and validation results. Section 4 presents test performance, error analysis, and interpretability findings. Section 5 concludes with limitations and future directions.

2. Methodology

This section presents the methodological pipeline adopted to address the task of Emotion Recognition in Conversation (ERC). We first describe the dataset and preprocessing steps used to structure the input at utterance and dialogue level. We then introduce the utterance-level representation strategies, based on frozen and fine-tuned Transformer embeddings, followed by the classification architectures designed to operate at utterance or dialogue level. Finally, we outline the training setup and evaluation protocol used to compare model configurations under consistent experimental conditions.

2.1. Dataset and Preprocessing

We base our experiments on the DailyDialog dataset (2), a high-quality collection of multi-turn dialogues designed for emotional and dialog act analysis. The dataset comprises approximately 13,000 dialogues, each annotated at the utterance

level with one of seven emotion labels (*neutral, happiness, sadness, anger, fear, disgust, surprise*) and one of four dialog acts (*inform, question, directive, commissive*). Dialogues typically contain between 5 and 15 turns, alternating between two speakers. Its structured, turn-based format and utterance-level emotion annotations make it suitable for evaluating context-aware modeling strategies.

To support different modeling strategies, we construct two variants of the dataset: an utterance-level version and a dialogue-level version. The utterance-level dataset is built by flattening each dialogue into independent (utterance, label) pairs, discarding turn order and conversational context. This format enables isolated classification, serving as a baseline for evaluating the added value of contextual information. In contrast, the dialogue-level dataset preserves the sequential structure of conversations. Each dialogue is represented as an ordered list of utterances paired with their corresponding emotion labels and speaker identifiers, enabling models to exploit contextual and temporal dependencies.

For each variant of the dataset, we apply consistent preprocessing steps. Raw dialogues are parsed from CSV files and converted into sequences of utterances, with associated emotion and dialog act labels parsed as lists. Turn-level speaker identities are assigned alternately (0 and 1) to reflect the two-party structure of the conversation. In the dialogue-level format, sequences are padded to the maximum dialogue length within each batch using standard PyTorch utilities, and labels are masked with -100 to exclude padding tokens from loss computation. We use the official train, validation, and test splits provided with the dataset, maintaining the original class distribution without rebalancing. This ensures that both modeling approaches are evaluated under aligned input conditions.

2.2. Embeddings

We compare three Transformer-based sentence embedding strategies to evaluate the impact of representation type on ERC performance. The first two rely on frozen encoders: RoBERTa-base and Sentence-BERT (all-mpnet-base-v2). For RoBERTa, we extract token-level representations and apply mean pooling over the final hidden states, weighted by the attention mask. Sentence-BERT directly outputs sentence-level embeddings optimized for semantic similarity (3). In both cases, the encoder weights remain fixed during training.

The third strategy involves fine-tuning the same RoBERTa-base model on the ERC task (4), using a classification head and cross-entropy loss. After training, the head is discarded, and utterance embeddings are extracted via the same mean pooling procedure. This setup allows us to assess whether task-adaptive representations offer performance advantages over general-purpose encodings, especially in combination with context-aware models.

2.3. Model Architectures

We explore two categories of classification models, corresponding to different levels of contextual awareness. The first

group includes utterance-level models that treat each input independently, relying solely on the semantic content of individual utterances. The second group consists of dialogue-level models that process entire conversations as sequences, enabling the integration of contextual and speaker-dependent information across turns. These architectures are designed to evaluate how performance varies as a function of access to dialog-level context and the ability to leverage context-sensitive representations.

Utterance-Level Models Utterance-level models operate under the assumption that each utterance carries sufficient semantic information to predict its emotional label without relying on dialogue context. We experiment with two variants. The first is a multilayer perceptron (MLP) classifier that maps each input vector to a seven-dimensional output via two hidden layers with ReLU activation and dropout. The second model consists of a RoBERTa-base encoder paired with a classification head, trained to predict the emotion label from the [CLS] token representation. This model operates at the utterance level, with each input processed independently using cross-entropy loss. The encoder is partially fine-tuned, with updates restricted to the last two layers during training.

Dialogue-Level Models Dialogue-level models aim to leverage the sequential and contextual structure of multi-turn conversations. We evaluate two architectures in this category. The first is a unidirectional GRU that processes each dialogue as a sequence of utterance embeddings (5). Turn-level speaker identities are embedded and concatenated to the input vectors before passing through the GRU layer. The output at each time step is fed to a linear classification layer to predict the corresponding emotion label. The second architecture replaces the GRU with a Transformer encoder (6). Each input sequence is augmented with speaker embeddings and positional encodings, projected to a fixed model dimension, and passed through multiple self-attention layers. A shared MLP classifier then maps the contextualized representations to emotion labels. Both models support variable-length input sequences through padding. The Transformer uses attention masks to ignore padding tokens, while both models apply label masking with -100 to exclude padded positions from the loss. This comparison allows us to assess whether attention-based models provide a meaningful advantage over recurrent architectures in modeling dialog-level emotional dependencies.

2.4. Training Setup and Evaluation Protocol

Evaluation Protocol Model performance is evaluated using macro-F1 and micro-F1 scores, both computed by excluding the *neutral* class. This choice reflects the label distribution of DailyDialog, where *neutral* dominates and carries limited discriminative value. Macro-F1 measures unweighted class-wise F1 and is appropriate under severe class imbalance. Micro-F1 aggregates predictions over non-neutral instances, capturing overall classification accuracy. These metrics are standard in ERC benchmarks on DailyDialog and are computed at the utterance level (7) (8) (9).

Training Setup All models are trained using the Adam optimizer with early stopping based on validation macro-F1 (ex-

cluding the *neutral* class). For the utterance-level MLP classifier, we use a learning rate of 1×10^{-3} and a batch size of 256. The RoBERTa-based classifier is partially fine-tuned by updating only the last two encoder layers and the classification head, with a learning rate of 1×10^{-4} . Input sequences are tokenized with a maximum length of 64, applying padding and truncation.

Dialogue-level models are trained on padded sequences with label masking (`ignore_index = -100`) to exclude padding positions from loss computation. The GRU model uses packed sequences and a dropout rate of 0.3. The Transformer model is trained with a learning rate scheduler combining linear warm-up for the first 200 steps and plateau-based decay (ReduceLROnPlateau) thereafter. Attention masks are applied to ignore padding during encoding. All models are trained for up to 80 epochs, and reproducibility is ensured by fixing random seeds across NumPy, PyTorch, and CUDA.

3. Experiments

This section presents the experimental design used to evaluate the proposed models on the DailyDialog dataset. We describe the configurations for training and validation, including the combinations of embeddings and classifiers explored. We also report the evaluation metrics adopted to compare model performance, with a focus on the impact of contextual modeling across different architectures.

3.1. Experimental Setup

We use the official train, validation, and test splits provided with the DailyDialog dataset, comprising 11,118, 1,000, and 1,000 dialogues, respectively. These splits preserve the original class distribution and are used consistently across all experiments.

Two architectural strategies are compared. The first is an end-to-end setup based on a RoBERTa-base encoder partially fine-tuned on the ERC task (last two layers), combined with a classification head trained to predict emotion labels directly from the [CLS] representation. This model operates at the utterance level, with each input processed independently.

The second is a modular pipeline, where utterance-level embeddings are precomputed and kept frozen during classifier training. We evaluate three types of sentence embeddings: (i) static contextual embeddings from frozen RoBERTa-base, (ii) task-adaptive embeddings obtained by discarding the classification head from the fine-tuned RoBERTa model, and (iii) static embeddings from Sentence-BERT (all-mpnet-base-v2), optimized for semantic similarity.

Each embedding type is tested with two levels of modeling: (a) a multilayer perceptron (MLP) for utterance-level classification, and (b) GRU or Transformer encoders for context-aware dialogue-level classification.

Model selection is based on validation macro-F1, computed by excluding the *neutral* class (label 0). Evaluation is performed at the utterance level. Both macro-F1 and micro-F1 scores are calculated over the non-neutral emotion classes {1, 2, 3, 4, 5, 6}. In dialogue-level models, padded positions are

excluded from loss computation and evaluation via label masking (`ignore_index = -100`).

All models are trained following the configuration described in Section 2.4.

3.2. Validation Results

We report validation performance for all model configurations, comparing macro-F1 and micro-F1 scores computed over non-neutral classes. As defined in Section 2.4, macro-F1 (excluding the *neutral* class) is used as the model selection criterion. Table 1 summarizes the results for each combination of embedding type and classification architecture.

Table 1: Validation performance (macro-F1 and micro-F1, excluding the *neutral* class) for each embedding-model combination.

Emb.	Model	Macro F1	Micro F1
–	RoBERTa + CLF (joint)	33.80	56.55
R	MLP	31.26	52.75
R-FT	MLP	28.98	55.42
SBERT	MLP	29.94	53.95
R	GRU	26.24	53.27
R-FT	GRU	32.32	55.56
SBERT	GRU	27.10	51.38
R	Transformer	31.59	54.37
R-FT	Transformer	38.09	55.60
SBERT	Transformer	29.95	53.34

Embeddings: Utterance representations from: R = frozen RoBERTa-base; R-FT = RoBERTa-base + CLF (joint) model, partially fine-tuned on ERC (last two layers), then frozen; SBERT = Sentence-BERT (all-mpnet-base-v2, frozen).

The best-performing configuration is the Transformer encoder operating on partially fine-tuned RoBERTa embeddings, which achieves the highest macro-F1 score (38.09). This confirms that combining task-adaptive representations with dialogue-level context modeling is the most effective strategy for capturing non-neutral emotional variation.

The RoBERTa-based utterance-level classifier (RoBERTa + CLF), which is partially fine-tuned end-to-end, obtains the second-best macro-F1 score (33.80). Despite lacking access to dialogue context, it outperforms all MLP and GRU models, even when they use the same fine-tuned embeddings as frozen input. This highlights the advantage of jointly adapting the encoder during training, rather than decoupling representation learning and classification.

The comparison between MLP and GRU reveals no consistent gain from sequence modeling. GRU slightly outperforms MLP when using R-FT embeddings (32.32 vs. 28.98), but performs worse with frozen RoBERTa and SBERT embeddings. These results suggest that GRU does not exploit dialogue-level dependencies more effectively than simpler feedforward architectures in this setting.

Across all configurations, models based on SBERT yield the lowest macro-F1 scores. This indicates that sentence embeddings optimized for semantic similarity are suboptimal for emo-

tion recognition, and reinforces the importance of task-specific adaptation.

Based on these results, we select the Transformer + R-FT configuration for final evaluation in Section 4.

4. Results

This section presents the evaluation of the best-performing model—Transformer with RoBERTa fine-tuned embeddings—on the test set. We report its overall performance, analyze typical classification errors, and discuss qualitative examples and interpretability aspects to better understand the model’s behavior.

4.1. Test Performance

We evaluate the selected model—a Transformer encoder operating over partially fine-tuned RoBERTa embeddings—on the held-out test set. Metrics are computed following the same protocol used during validation: macro-F1 and micro-F1 scores are calculated at the utterance level, excluding the *neutral* class.

The model achieves a macro-F1 of 44.94 and a micro-F1 of 54.54 on the non-neutral emotion classes. These results indicate consistent generalization across data splits and show that the model maintains performance on previously unseen dialogues from the same domain. Compared to the validation macro-F1 (38.09), the test performance improves by +6.85 points.

In the context of published benchmarks on DailyDialog, our model remains below the best-performing approaches. For instance, EmoOne-RoBERTa achieves 61.67 micro-F1 and 55.84 macro-F1 (8), while EmotionIC reports 60.13 and 54.19 (9), respectively. Despite this gap, our approach uses a simplified architecture and limited fine-tuning, demonstrating that a reduced configuration can still achieve reasonable performance without relying on task-specific enhancements.

Table 2: Test performance of the selected model (Transformer + R-FT) on non-neutral emotion classes.

Metric	Score
Macro-F1 (no neutral)	44.94
Micro-F1 (no neutral)	54.54

Table 3 presents precision, recall, F1 score, and support for each non-neutral emotion class in the test set. The model performs best on *happiness* (F1 = 60.69), which is also the most frequent non-neutral label. This suggests that exposure during training contributes significantly to generalization. *Fear*, despite being the sparsest class (17 instances), achieves the second-highest F1 (53.33), driven by high precision (61.54) and moderate recall (47.06), indicating that predictions are accurate when made, though limited in coverage.

Performance drops markedly on *sadness* (F1 = 32.20), *disgust* (36.17), and *anger* (39.62), which all have limited support and exhibit balanced but low precision and recall. Notably, *surprise* achieves 47.62 F1 with 116 examples, showing

higher recall (51.72) than precision (44.12), indicating a tendency toward overprediction on ambiguous cases. This highlights that frequency is a necessary but not sufficient condition for improved classification, as semantic ambiguity and lexical overlap remain limiting factors.

These results indicate that both class imbalance and label sparsity affect performance, but lexical distinctiveness and semantic overlap between emotions are also key factors. A more detailed examination of misclassifications is provided in Section 4.2.

Table 3: Per-class precision, recall, F1 score, and support (non-neutral emotions).

Class	Prec.	Rec.	F1	Support
Happiness	68.52	54.47	60.69	1019
Surprise	44.12	51.72	47.62	116
Fear	61.54	47.06	53.33	17
Anger	44.68	35.59	39.62	118
Disgust	36.17	36.17	36.17	47
Sadness	32.04	32.35	32.20	102

4.2. Error Analysis

Figure 1 reports the confusion matrix computed on the test set and normalized over true labels. All emotion classes are included. Although macro-F1 and micro-F1 scores are calculated excluding the *neutral* class, the matrix reflects all predictions, including misclassifications into *neutral*.

The model exhibits a strong tendency to predict *neutral* for low-frequency emotions. For example, 53 of true *anger* and 62 of true *sadness* instances are misclassified as *neutral*, exceeding their correct classification rates. This pattern is consistent for *fear* (47 to *neutral*) and *disgust* (51), indicating a fallback behavior under uncertainty or class imbalance.

Among non-neutral confusion, *anger* and *disgust* are frequently misclassified into one another (6 and 11, respectively), as are *surprise* and *happiness* (8). These patterns suggest overlap between emotions with similar valence, likely driven by shared lexical cues and contextual ambiguity in expression. Such confusions highlight the challenge of distinguishing emotions that co-occur in discourse or share common surface forms, particularly when contextual disambiguation is limited or underutilized by the model.

The diagonal remains dominant for *happiness* and *surprise*, which retain 54 and 52 recall respectively, confirming partial robustness on more frequent emotions. However, the overall error distribution reveals that many emotionally labeled utterances are absorbed into the *neutral* class, highlighting a structural limitation in distinguishing infrequent emotional classes when *neutral* acts as a dominant fallback.

4.3. Interpretability

We analyze how the final model encodes and utilizes emotional signals by examining its internal representations and attribution patterns. Specifically, we investigate (i) the geome-

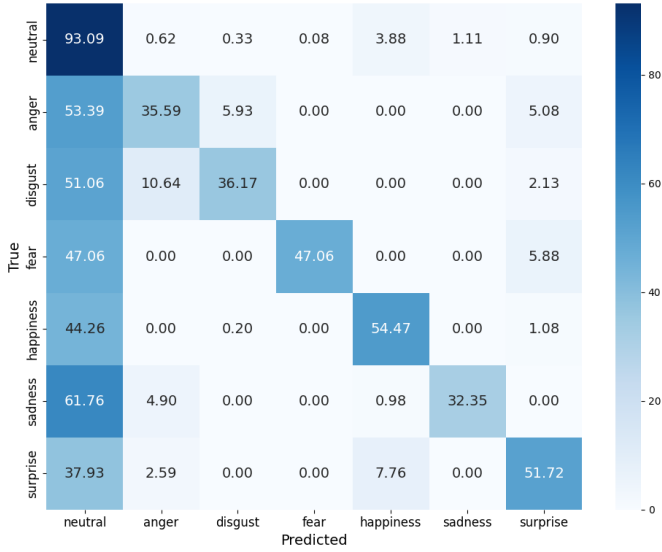


Figure 1: Confusion matrix normalized over true labels (test set, all classes included).

try of the fine-tuned RoBERTa embedding space, (ii) the attention distribution across dialogue turns in the Transformer encoder, and (iii) Integrated Gradients (IG) attribution scores, which provide a gradient-based estimate of context contribution to each prediction. These analyses aim to determine whether the model’s decision process aligns with emotion-specific structure and whether context is effectively integrated at inference time.

Embedding space visualization To investigate how the model represents emotional information, we visualize the utterance-level embeddings obtained from the fine-tuned RoBERTa encoder used in the final configuration. Figure 2 shows a two-dimensional t-SNE projection of the test set embeddings, color-coded by true emotion labels.

The plot reveals partial clustering of emotion classes. *Happiness* forms a dense and well-separated region in the lower-right area, while *surprise* is also visibly concentrated in the upper-left. Other classes such as *anger*, *disgust*, and *fear* are distributed in overlapping regions with no clear boundaries. *Sadness* appears semi-clustered, but still entangled with other negative-valence emotions. Notably, *neutral* instances dominate the left side of the plot and are largely disjoint from the emotional classes.

These patterns suggest that the fine-tuned encoder captures some degree of emotion-specific structure, especially for distinguishable high-frequency classes. However, significant overlap remains among negative emotions, indicating that class separation is not fully resolved at the representation level. The overlapping of negative-valence emotions in the embedding space aligns with the confusion patterns observed in Section 4.2, suggesting that representational similarity contributes to systematic misclassification. As t-SNE preserves local but not global distances (10), the observed clusters provide qualitative—but not strictly quantitative—evidence of representational separability.

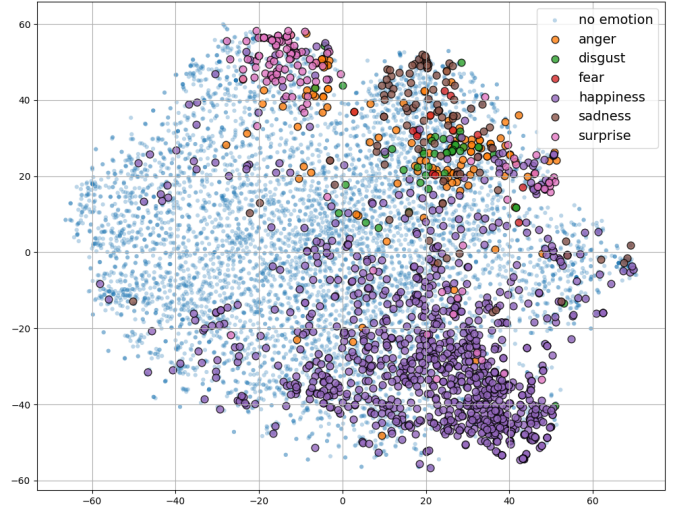


Figure 2: t-SNE projection of fine-tuned RoBERTa embeddings on the test set, colored by true emotion labels.

Dialog-level attention analysis To analyze how the Transformer encoder integrates contextual information across turns, we examine the average attention assigned to utterances at different relative positions. Figure 3 shows the distribution of attention weights with respect to the distance $i - j$ between the query utterance i and each context position j , grouped by the predicted emotion of the query utterance.

Anger, *disgust*, and *fear* exhibit strong self-attention peaks (values at $i = j$) as well as high attention on positions ± 2 , which correspond to turns spoken by the same speaker due to the alternating structure of the dialogues. This suggests that the model relies on intra-speaker consistency to identify these emotions, focusing on local patterns within a speaker’s trajectory.

Happiness behaves differently, assigning relatively low attention to the utterance itself (8.67), markedly lower than for other emotions, and distributing focus more evenly across the surrounding turns ($\pm 1, \pm 2$). This indicates a higher dependence on immediate dialogic context for its recognition, rather than self-contained emotional signals.

Sadness and *surprise* combine both strategies: while self-attention remains dominant, the attention weights on surrounding turns are more balanced. These emotions are influenced by both the content of the utterance itself and the local conversational flow.

These findings suggest that emotion-specific attention patterns emerge from the interaction between speaker identity and dialogic structure, and that the Transformer adapts its focus depending on the emotion being predicted.

Integrated Gradients-based context attribution

To complement the structural insights derived from attention weights, we apply *Integrated Gradients* (IG) to quantify the contribution of each context utterance to the model’s decision. Unlike attention, IG provides a causal measure of input relevance by integrating gradients along a path from a baseline to the actual input (11), offering a more faithful estimate of feature attribution (12).

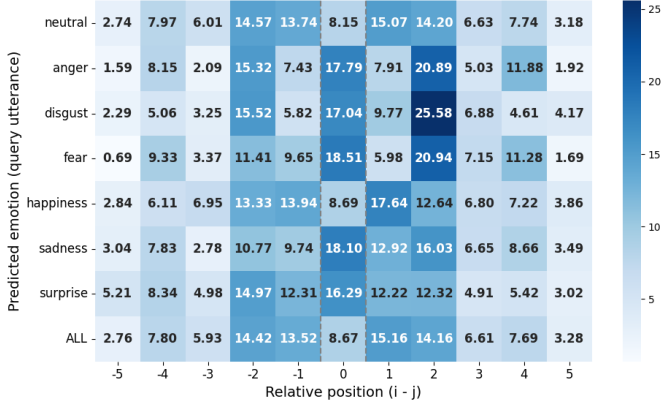


Figure 3: Attention distribution by relative position ($i - j$) for each predicted emotion. Values are averaged across all attention heads and layers.

We aggregate IG scores by relative position ($i - j$) between the query utterance i and each context position j , normalize them row-wise to sum to 100, and group results by predicted emotion class (Figure 4).

All emotions show a dominant peak at the query position ($i = j$), confirming strong self-attribution. However, the degree of contextual reliance varies across classes:

Surprise and *fear* exhibit the highest self-attribution scores (63.7 and 52.5, respectively), indicating minimal dependence on surrounding context.

Anger and *disgust* show secondary peaks at positions ± 2 , corresponding to turns from the same speaker, suggesting reliance on intra-speaker context.

Happiness and *sadness* display broader attribution spread across positions ± 1 and ± 2 , involving both speakers. This reflects greater sensitivity to short-range conversational flow rather than strictly self-contained emotional signals.

These findings reveal emotion-specific attribution patterns, where contextual integration depends both on emotion type and speaker dynamics.

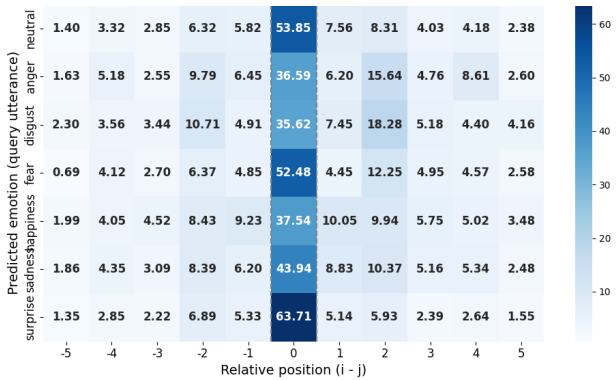


Figure 4: Row-normalized Integrated Gradients by relative distance ($i - j$) for each predicted emotion. Values indicate the average contribution of each context utterance to the predicted label.

4.4. Qualitative Examples

We report qualitative examples from the test set to analyze typical prediction behaviors. Each example includes the full dialogue, with true and predicted emotion labels for each utterance.

The target utterance is highlighted in bold. For that turn, we report the attention weights from the Transformer encoder and the Integrated Gradients (IG) attribution scores across all utterances in the dialogue. Attention reflects the distribution of representational focus during encoding; IG quantifies the contribution of each input to the final prediction. All values are normalized within each example.

Example 1 – Anger consistently misclassified as neutral

- 0 John, I was looking through some magazines for ideas about where we might go on vacation this year.
[true: neutral, pred: neutral]
Attn: 3.07 — IG: 1.00
- 1 I've already told my buddy, Mark, that I am going hunting with him in Alaska. [true: neutral, pred: neutral]
Attn: 5.22 — IG: 1.17
- 2 You can't be serious! [true: anger, pred: surprise]
Attn: 18.15 — IG: 5.13
- 3 Hey, I've always gone hunting or fishing on vacation. I am sorry that bothers you. [true: anger, pred: neutral]
Attn: 22.66 — IG: 6.76
- 4 After a year together, I thought it pretty safe to assume that we would automatically spend our vacation together. [true: anger, pred: neutral]
Attn: 5.91 — IG: 5.39
- 5 **Says who? I don't think that is necessarily the case.** [true: anger, pred: neutral]
Attn: 2.81 — IG: 44.37
- 6 You know, now that I think about it, I really don't have much more to say to you at all! [true: anger, pred: neutral]
Attn: 9.02 — IG: 4.62
- 7 Whatever you say! [true: anger, pred: neutral]
Attn: 33.17 — IG: 5.80

Turn 5 occurs within a contiguous sequence of utterances labeled as *anger*, yet the model outputs *neutral* for all of them, except for Turn 2, which is incorrectly classified as *surprise*. Attention is distributed across multiple turns, with local maxima at Turn 7 (33.17), Turn 3 (22.66), and Turn 2 (18.15), indicating a diffuse focus over the dialogue context. However, the IG attribution for Turn 5 is dominated by the self-utterance (44.37), suggesting that the classification decision is driven almost entirely by local lexical features. The low contextual attribution, despite surrounding emotionally coherent turns, indicates that the model fails to exploit temporal emotion progression and intra-speaker consistency.

Example 2 – Anger misclassified as disgust

- 0 Hey Sandra, what's wrong? You look furious! [true: neutral, pred: neutral]
Attn: 3.67 — IG: 2.04
- 1 I've had it with Tim! I think I'm going to quit! [true: anger, pred: anger]
Attn: 43.19 — IG: 16.45
- 2 But Sandra... you love your job! You spent months trying to get this position. It's perfect for you! [true: neutral, pred: neutral]
Attn: 4.31 — IG: 3.67
- 3 **Yeah... I know. But I'm sick and tired of all the office power struggles!** [true: anger, pred: disgust]
Attn: 10.89 — IG: 46.39
- 4 What kind of power struggles are you talking about? [true: neutral, pred: neutral]
Attn: 0.72 — IG: 2.15
- 5 Basically, I think Tim is threatened by me. I'm doing a better job than he is so he's trying to sabotage my career! [true: anger, pred: neutral]
Attn: 15.87 — IG: 12.18
- 6 So he talks about you behind your back? [true: neutral, pred: neutral]
Attn: 0.56 — IG: 1.79
- 7 Yes. He's all nice and friendly to my face. But the second I turn around he stabs me in the back! [true: anger, pred: anger]
Attn: 5.62 — IG: 1.44
- 8 I had a worker like that once. She was always taking credit for my ideas. [true: neutral, pred: neutral]
Attn: 0.35 — IG: 1.64
- 9 I should keep a better record of my correspondence with the clients so if necessary I can prove it was my effort. [true: neutral, pred: neutral]
Attn: 14.61 — IG: 11.70
- 10 My co-worker had a whole bunch of bad office habits. She liked to name-drop... we would be in a meeting and she would say, yesterday I was chatting with Tom. She meant Tom Solomon... our CEO. [true: neutral, pred: neutral]
Attn: 0.22 — IG: 0.53

Turn 3 is misclassified as *disgust* despite being labeled as *anger*. The utterance contains lexical cues that can plausibly indicate either class, increasing ambiguity. Attention is concentrated on Turn 1 (43.19) and Turn 5 (15.87), both produced by the same speaker, indicating that the model captures intra-speaker context during encoding. However, IG attribution is dominated by the target utterance (46.39), with secondary contributions from Turn 1 (16.45), Turn 5 (12.18), and Turn 9 (11.70). This suggests that, although context is integrated into the representation, the classification remains primarily driven by local lexical features. The model fails to disambiguate semantically overlapping emotions, relying heavily on surface forms without sufficient contextual disambiguation.

Example 3 – Fear correctly predicted

- 0 How's the bride-to-be? [true: neutral, pred: neutral]
Attn: 6.19 — IG: 3.56
- 1 Tired and stressed. This wedding is giving me a headache. [true: sadness, pred: neutral]
Attn: 35.41 — IG: 15.29
- 2 Just relax. With me as your maid of honor, everything will be fine! [true: happiness, pred: neutral]
Attn: 3.47 — IG: 3.10
- 3 **I'm worried that everything will be chaotic!** [true: fear, pred: fear]
Attn: 28.55 — IG: 58.73
- 4 Hey! I'm the tour guide. I'll take care of the Taiwanese guests. [true: neutral, pred: neutral]
Attn: 4.32 — IG: 5.22
- 5 You're going to have to tell them what to do in church, what everything means... [true: neutral, pred: neutral]
Attn: 11.70 — IG: 8.43
- 6 I will. Don't worry! [true: happiness, pred: neutral]
Attn: 10.35 — IG: 5.66

Turn 3 is correctly classified as *fear*, with attribution dominated by the self-utterance (IG: 58.73), indicating that the prediction is largely driven by explicit lexical indicators. Attention, however, peaks at Turn 1 (35.41), suggesting that the encoder incorporates prior emotional context into the intermediate representation. Turn 1 also contributes non-negligibly in IG (15.29), supporting the hypothesis that the model leverages upstream emotional cues. Turn 5 shows moderate impact in both attention (11.70) and IG (8.43), indicating that context plays a complementary role in supporting the prediction without overriding the local signal.

Example 4 – Context-dependent misclassification

- 7 Do you really have all of these drugs? Where do you get them from? [true: neutral, pred: neutral]
Attn: 1.80 — IG: 2.10
- 8 I got my connections! Just tell me what you want and I'll even give you one ounce for free. [true: neutral, pred: neutral]
Attn: 6.35 — IG: 5.45
- 9 Sounds good! Let's see, I want. [true: neutral, pred: neutral]
Attn: 6.82 — IG: 4.60
- 10 **Yeah?** [true: fear, pred: surprise]
Attn: 1.70 — IG: 64.78
- 11 I want you to put your hands behind your head! You are under arrest! [true: neutral, pred: neutral]
Attn: 62.46 — IG: 12.95

Turn 10 is misclassified as *surprise* instead of *fear*. The utterance is lexically underspecified, and its correct interpretation relies on future context—specifically Turn 11, which discloses the speaker's intent ("You are under arrest"). During encoding, the model attends primarily to Turn 11 (62.46), indicating that contextual signals are accessible. However, IG attribution is dominated by the target utterance (64.78), while Turn

11 contributes only marginally (12.95). This suggests a disconnect between representational focus and attribution, where the model encodes context but does not propagate it effectively into the classification decision. The result is a prediction driven by surface features of an ambiguous utterance, without sufficient contextual disambiguation.

5. Conclusions

This study assessed two architectural strategies for Emotion Recognition in Conversation: an end-to-end model with partial fine-tuning at the utterance level, and a modular pipeline with frozen embeddings used by utterance- and dialogue-level classifiers. The modular approach was tested with frozen and task-adaptive RoBERTa embeddings, as well as Sentence-BERT representations.

Results show that fine-tuning improves performance only when combined with dialogue-level modeling. The best configuration—a Transformer encoder operating on frozen embeddings extracted from the end-to-end RoBERTa classifier—outperforms all other models. The end-to-end model itself ranks second, confirming the benefit of joint adaptation even without context modeling. In contrast, utterance-level models using frozen embeddings perform consistently worse, and Sentence-BERT representations yield the lowest scores overall.

Interpretability analysis indicates that context is not always effectively propagated from representation to decision. Attention weights and gradient-based attribution both reveal emotion-specific patterns but also highlight a gap between contextual encoding and final prediction.

Future work could explore several directions. First, incorporating dialog act labels as auxiliary inputs may help disambiguate emotions with overlapping lexical patterns. Second, class imbalance in DailyDialog calls for targeted augmentation methods, such as controlled paraphrasing or generation techniques for low-resource emotions. Third, end-to-end training of RoBERTa and the dialogue-level Transformer—with either full or partial encoder updates—represents a promising direction. Joint fine-tuning of the two components may improve the alignment between utterance-level representations and dialogue-level context modeling, overcoming the limitations observed with frozen embeddings.

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